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Strategic Rail Network Optimization Model for Freight Transportation

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An optimization model for a rail freight network was developed as a support tool for planning and policy decisions involved in the improvement of rail networks on a regional and national level. The model was based on a strategic traffic assignment model, which was designed to model macronetworks with a high aggregation level, and was exclusively designed for freight traffic. The model contemplates road and rail transport modes and considers two types of cargo: intermodal cargo, which is generally transported in containers and is easily interchanged between different modes at intermodal terminals, and general cargo, which represents all the remaining cargo. The optimization process was based on a local search heuristic that delivers good solutions in a reasonable computing time. The quality of each network improvement solution was assessed on the basis of the reductions in total generalized costs and carbon dioxide emissions. This optimization model for a freight network is innovative because it is not limited: both the improvement of existing links and the construction of new ones are allowed, and the model does not have a limit on the number or variety of possibilities for network improvement. The adaptability to different conditions was emphasized when the model was applied to two networks under different investment scenarios because the model delivered considerably different solutions adapted to the conditions of each scenario.

Although freight transportation is an activity that plays a crucial role in the everyday life of any modern economy, being critical to a large part of the economy, it usually gets less attention in academia than its counterpart, passenger transportation. This is probably justified not only by the fact that it is not as appealing to policy makers and the general public as passenger transportation but also because it is a considerably more complex subject because of the multiplicity of goods transported, the complexity of the freight supply chain, and the difficulty with the collection of the needed data.

Although freight transportation may be less appealing and more complex than passenger transportation, it is important to study freight transportation through the use of models specifically made for such studies to account for its distinct characteristics and for the fact that the network investments needed to improve freight transportation can be considerably different from those needed to improve passenger transportation. For this reason, the network optimization model and the associated traffic assignment model that are presented in this paper have been developed specifically for this type of transportation, although they may be combined with passenger transportation models in the future to create a model for the whole transportation system.

The model presented here, developed in the scope of a broader project (*1*), uses a strategic planning traffic assignment model designed to model macronetworks with a high level of aggregation (*2*). This assignment model does not require detailed data inputs; and the outcome of its application is an estimate of the movement of freight at a regional, national, or international scale. It considers road and rail transport modes and is intended to simulate medium- and long-distance flows of inland transportation. The model contemplates two types of cargo, namely, (*a*) general cargo and intermodal cargo, to make a distinction between the cargo that may be easily interchanged between different modes at intermodal terminals, which is generally transported in containers, and (*b*) all remaining cargo. The characteristics described above make this traffic assignment model particularly suited for the planning and policy decisions that will be performed by the network optimization model (*3*).

The optimization process itself is flexible and innovative and allows upgrades to both the quality of existing rail and intermodal terminal links and the construction of new ones, with no limit on the number or variety of improvement solutions. This possibility for unlimited improvements is achieved by definition of a set of possible link levels for each link type, according to the users' preferences, including the mere possibility of building of a link. The quality of each network improvement solution is assessed on the basis of the reduction of total generalized costs and carbon dioxide (CO_2) emissions, with the weights given to each of those parameters defined by the user according to the user's preferences. The optimization model is based on a search heuristic and tries to meet a balance between efficiency and effectiveness by delivery of good solutions in a reasonable computing time.

This paper is structured in six sections. After this introduction, a background section contains a brief literature review on the subject of freight traffic assignment and network optimization models. The following sections are dedicated to the traffic assignment model and the network optimization process. An application of the network optimization model is then described, and the last section is dedicated to the final conclusions.

BACKGROUND

Models for Freight Traffic Assignment

The different traffic assignment techniques that are presented in the literature can be divided into four groups: all-or-nothing, equilibrium, stochastic–multiflow, and stochastic–equilibrium (*4*). The two factors that determine to which of the four groups a model belongs are capacity constraints imposed by congestion (equilibrium and

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stochastic–equilibrium models) and a variable perception of costs (stochastic–multiflow and stochastic–equilibrium models). Capacity constraint models are those in which the capacity of links is limited and time penalties because of congestion may be included when certain traffic limits are exceeded. The variable perception of costs reflects whether the mode and route choice decisions are made uniquely on the basis of the lowest generalized cost or if some stochasticity is included, in which stochasticity reflects a variable perception of costs and, consequently, spreading of the traffic through different modes and routes.

The assignment techniques most commonly used for freight traffic assignment models are the all-or-nothing and the equilibrium techniques, with stochastic–multiflow models seldom being used. According to the authors' best knowledge, the use of stochastic–equilibrium models in this area of study has been limited, probably because of their complexity.

Although the literature describes many different models, with many being created for just one specific work, two major freight traffic assignment models are worth mentioning because of their importance and extensive use. Those are STAN, which was developed in 1990 in Canada (*5, 6*) and which makes use of an equilibrium assignment technique, and NODUS software, which was developed in Belgium a few years later (*4, 7, 8*) and which has been used with the three most common assignment techniques: the all-or-nothing, equilibrium, and stochastic–multiflow techniques. The fact that various techniques were used with NODUS software shows that each one has its own advantages, which should be considered when the type of technique to be used is chosen.

An all-or-nothing technique is the most straightforward and is the best suited for cases in which other assignment techniques are considered to be too complex or simply not fit for the proposed approach (*7, 8*). The equilibrium assignment technique is best suited for congested networks. In this technique, the effects of congestion are taken into account by the use of cost penalties when traffic is close to capacity (*5, 6*). The stochastic–multiflow technique distributes traffic by different possible routes, which ensures that the path with the least generalized costs never receives all the traffic. This is a valuable feature when one is dealing with strategic aggregated models in which generalized transportation costs are just an estimate of the average costs and which are not able to incorporate many factors that have a decisive influence on the modal or route choice, such as shipment size (*9*), frequency of service (*10*), service quality (*11*), and existence of an integrated door-to-door logistic chain (*12*). The method used to distribute traffic is the logit formulation, which has consistently been chosen to address this problem because of its versatility and convenience (*4, 13, 14*).

Models for Network Optimization

The aim of a network optimization model is to find out how the transportation network should be improved to minimize or maximize (as applicable) the appropriate indicators, defined by the user. The optimization parameters can vary from case to case and must have an associated quantitative indicator for them to be objectively analyzed. Many parameters can be used to assess the quality of each solution, namely, the total generalized cost, the robustness of the network under uncertain conditions (*15*), and the environmental impact (*16*).

The network improvement operations that may be performed to optimize the network can be improvements to the quality of existing links or the construction of new links from scratch, with each specific improvement operation having an associated investment cost to quantify the investment made in a given network improvement scenario. The availability of well-defined network conditions and improvement possibilities, as well as the corresponding costs, is the basic ingredient needed for the development of a network optimization process able to optimize the way in which investments are made to minimize or maximize selected parameters.

Most of the research on the subject of network optimization found in the literature was performed by the use of two types of models: the discrete network design problem (*16–18*) and the continuous network design problem (*19*). Although the former tends to concentrate on the addition of new links and the latter tends to concentrate on the (continuous) improvement of existing links, it is also possible to use a discrete approach that allows both the addition of new links and the improvement of existing links (*16*).

Because of the considerable complexity of the transportation networks and the discrete nature of most models, no practical analytical solution for this problem exists, a situation that leads to the adoption of heuristic techniques. Several techniques have successfully been used to address this kind of problem. The predominant ones are metaheuristic techniques, such as tabu search, simulated annealing, ant colony optimization, and genetic algorithms (*16–18, 20, 21*).

Although various network optimization models are specifically made for freight transportation, they tend to focus on the optimization of specific equipment, such as intermodal terminals (*18*), or to be limited either because they allow only the creation of new links or the improvement of new links or because they have a limited search space with only a limited number of improvement possibilities (*17*).

MODEL FOR FREIGHT TRAFFIC ASSIGNMENT

General Attributes of the Model and Computation of Shortest Paths

The model presented here is a strategic freight traffic assignment model that is based on a previous model developed by the authors (*22*) that was upgraded to allow the introduction of separate link characteristics for each type of cargo and for the introduction of congestion in rail nodes. It considers two types of cargo: general cargo and intermodal cargo. The latter represents the share of freight that can be easily transferred between the road and rail modes at intermodal terminals, namely, containerized cargo, with general cargo representing all other cargo.

The model was developed for the assignment of traffic on road and rail networks, but apart from the links representing those networks, it also includes other types of complementary links. Those links include intermodal terminals that establish the connection between the road and rail networks or between centroids and the rail network, as well as virtual links that are included to represent congested rail nodes, which are an important limitation to the capacity of rail networks. The model also includes connector links, which make the connection between the road network and the centroids that represent the traffic-generating regions. Each link has a set of attributes, with some of them being inherent to the link, namely, its length and capacity, and with others depending on the vehicles that use those links, which may be different for each type of cargo. Those attributes are average speed, capacity, cost per distance, and CO_2 emissions for the vehicles and the value of time for each type of cargo.

The generalized cost per unit of cargo is calculated by use of a vehicle cost component and a time cost component and is given by Equation 1:

$$p_i^k = \frac{l_i * q_i^k}{r_i^k} + \frac{l_i}{s_i^k} * v_i^k \qquad \forall i \in I, k \in J \tag{1}$$

where

I = set of all links;
J = {Gen, Int}, where Gen is general cargo and Int is intermodal cargo;
p_i^k = generalized cost per unit of cargo on link i for cargo type k;
l_i = length of link i;
q_i^k = vehicle cost per distance on link i for cargo type k;
r_i^k = vehicle capacity on link i for cargo type k;
s_i^k = vehicle average speed on link i for cargo type k; and
v_i^k = value of time on link i for cargo type k.

On the basis of the defined generalized costs, it is possible to calculate the shortest path (with the least generalized cost) between any given pair of nodes for each type of cargo. The shortest-path algorithm used in the model is the Floyd–Warshall algorithm with path reconstruction, which computes the value of the shortest paths between all the nodes, as well as the path itself [the links used in each shortest path (*23*)].

Assignment Process

With the basic features of all of the models properly defined, it is now possible to describe the assignment process, whose main features are presented in Figure 1.

As Figure 1 shows, different assignment techniques may be used for the two types of cargo, mainly because intermodal cargo is allowed to use only intermodal terminals to transfer between different transport modes; this technique is a solution already applied to other studies in this area (*8*). Although intermodal cargo may use more than one mode of transport per trip, general cargo is limited to the use of the same transport mode in each trip. This restriction is reflected in the traffic distribution techniques that are used for each type of cargo.

In the case of general cargo, as each trip may use only one mode of transport, a clear decision about the least-costly mode between road and rail transport must be made. This allows the distribution of traffic between the two modes through the use of a stochastic–multiflow technique, which is implemented by use of a logit function (*14*), which gives the percentage of traffic that uses each mode. Intermodal cargo traffic is assigned to the least-costly path between the origin and the destination, which may include both road and rail transport modes. This assignment is made through the use of an all-or-nothing technique, which allows road and rail links to be used indifferently on every trip, as long as they are part of the absolute least-expensive path.

Although the model does not consider capacity limits on road links because congestion is mostly observed in and around urban areas (*4*) and not on intercity routes, the model considers capacity limits on rail links and at intermodal terminals. This consideration is justified by the fact that the capacity of rail links and intermodal terminals is relatively rigid, with empirical evidence showing that rail links are more likely to be used to capacity than their modern intercity road link counterparts. Given that most of the capacity problems in rail networks are due to specific points in the network, such as congested rail junctions, the model allows the introduction of congested rail nodes with a limited capacity. As shown in Figure 2,

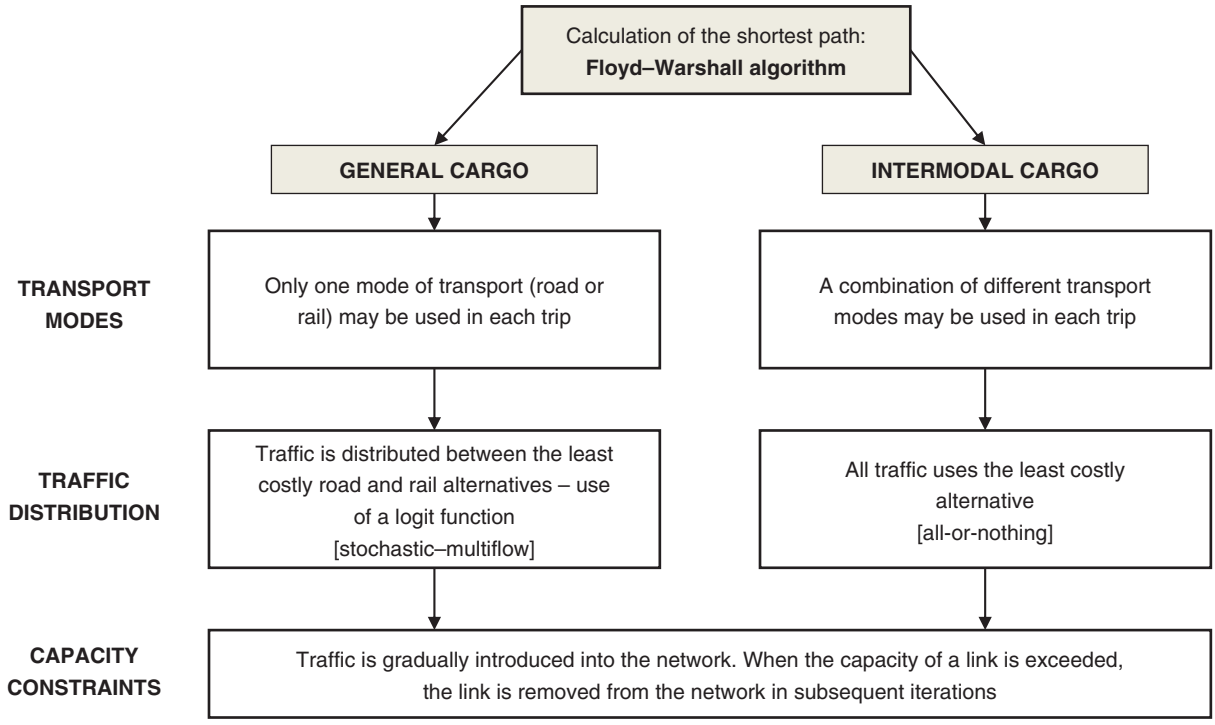


FIGURE 1 Assignment process.

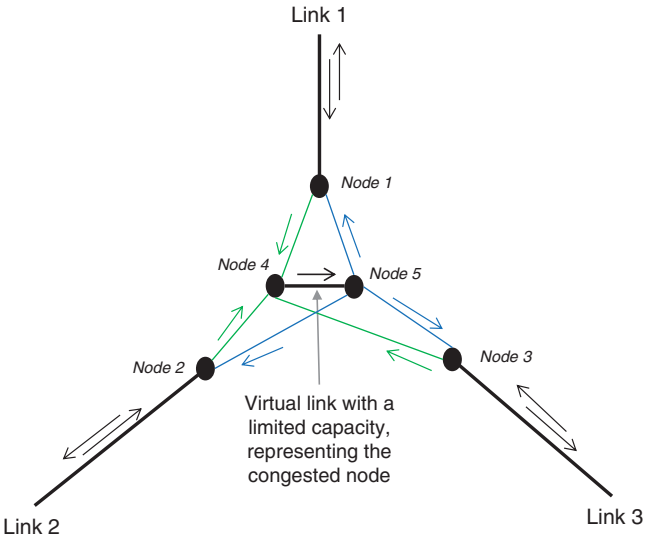


FIGURE 2 Congested rail nodes scheme showing an example of congested rail node with three rail links converging on it.

which displays the technique used to model rail nodes (*5*), the model considers a virtual link that represents the total capacity of the rail node.

The physical capacity of each link is defined in the network, which means that the capacity that remains for freight trains is the total capacity minus the flow of passenger trains. That flow is obtained by assignment of an origin–destination matrix of passenger trains to the rail network through the use of the shortest-distance path.

Although the model includes capacity constraints, this is not done by use of an equilibrium model. The total freight flow is gradually inserted into the network, with the user defining the number of interactions in which traffic flow is introduced into the network. If any new link has reached its capacity after each iteration, it is removed from the network and the shortest paths and traffic distributions are recalculated. This process continues until all the traffic is assigned to the network.

MODEL FOR NETWORK OPTIMIZATION

Overview of Optimization Process

The first step in the development of a network optimization process is definition of the network structure adopted, with all possible link levels and network improvement possibilities being defined. The solution used in this model is a network structure in which the links have a limited number of discrete quality levels and in which each level corresponds to a different link type.

The quality of a link can vary from zero, which corresponds to the mere possibility that a link will be built, to the highest level, which corresponds to the best possible link quality. Each link level has an associated set of characteristics for each type of cargo that may be freely defined by the user.

This network structure allows both improvements to links and the construction of new links. These conditions allow unlimited possibilities for improvement. The model allows the improvement of rail, intermodal terminal, and virtual links, which means that all the links that are related to rail transport may be improved to meet the goal of the model, which is the optimization of rail networks. All possible improvement operations that are defined by the user must have an associated cost to quantify the amount of money that will be spent in each improvement scenario.

The factors that are considered for the assessment of the quality of each network improvement solution are the total generalized costs and the total emissions of CO_2. The total generalized cost reflects the economic costs that are supported by the freight carriers and that are used to make transportation decisions. The total emissions of CO_2 quantify the total amount of CO_2 emitted by all vehicles transporting freight and serve as a measure of the environmental impact caused by freight transportation. The quality of each improvement solution is measured on the basis of the minimization of both of these parameters. Given the existence of more than one optimization parameter, it is necessary to define the weight that will be given to each one of them, which is something that is defined by the user according to the planning priorities of each case.

The mathematical formulation of this problem can be described as follows. The objective is

$$\min N = \frac{w_{GC}}{w_{GC} + w_{EI}} * \frac{GC - GC_0}{GC_0} + \frac{w_{EI}}{w_{GC} + w_{EI}} * \frac{EI - EI_0}{EI_0} \tag{2}$$

subject to

$$\sum_{i \in I} m_i \le B \qquad i \in I \tag{3}$$

where

$$GC = \sum_{i \in I} \sum_{k \in J} f_i^k * r_i^k * p_i^k \qquad \forall i \in I, k \in J \tag{4}$$

$$EI = \sum_{i \in I} \sum_{k \in J} f_i^k * c_i^k \qquad \forall i \in I, k \in J \tag{5}$$

$$m_i = \begin{cases} 0 & \text{if } a_i \in \{0,1\} \quad \forall i \in I \quad (6) \\ l_i * d_{a_i u_i z_i} & \text{if } a_i \in \{2\} \quad \forall i \in I \quad (7) \\ d_{a_i u_i z_i} & \text{if } a_i \in \{3,4\} \quad \forall i \in I \quad (8) \end{cases}$$

The potential link status for each link type is

$$u_i \in \{1,2,3,4,5\} \text{ and } z_i \in \{1,2,3,4,5\} \qquad \text{if } a_i \in \{0\} \qquad \forall i \in I \tag{9}$$

$$u_i \in \{1\} \text{ and } z_i \in \{1\} \qquad \text{if } a_i \in \{1\} \qquad \forall i \in I \tag{10}$$

$$u_i \in \{0,1,2,3,4,5\} \text{ and } z_i \in \{0,1,2,3,4,5\} \qquad \text{if } a_i \in \{2\} \qquad \forall i \in I \tag{11}$$

$$u_i \in \{0,1,2,3,4\} \text{ and } z_i \in \{0,1,2,3,4\} \qquad \text{if } a_i \in \{3\} \qquad \forall i \in I \tag{12}$$

$$u_i \in \{1,2,3,4,5,6\} \text{ and } z_i \in \{1,2,3,4,5,6\} \qquad \text{if } a_i \in \{4\} \qquad \forall i \in I \tag{13}$$

The possible link improvement operations for each link type are

$$z_i = u_i \quad \text{if } a_i \in \{0,1\} \quad \forall i \in I \tag{14}$$

$$z_i \geq u_i \quad \text{if } a_i \in \{2\} \text{ and } u_i \in \{0,1,2,3,4,5\} \quad \forall i \in I \tag{15}$$

$$z_i \in \{3,5\} \quad \text{if } a_i \in \{2\} \text{ and } u_i \in \{3\} \quad \forall i \in I \tag{16}$$

$$z_i \geq u_i \quad \text{if } a_i \in \{3,4\} \quad \forall i \in I \tag{17}$$

where

N = weighted change in total generalized cost and environmental impact,
GC = total generalized cost,
GC_0 = initial total generalized cost,
EI = total environmental impact,
EI_0 = initial total environmental impact,
w_{GC} = weight given to generalized costs,
w_{EI} = weight given to environmental impact,
f_i^k = traffic flow (both ways) on link i for cargo type k,
c_i^k = CO_2 emissions per vehicle per kilometer on link i for cargo type k,
B = total available budget,
m_i = investment on link i,
a_i = link type of link i,
u_i = original link status of link i,
z_i = new link status of link i, and
$d_{a_i u_i z_i}$ = unit investment to upgrade link type a from original link status u to new link status z.

Except for small networks, this network optimization process is extremely difficult to solve to exact optimality by use of an analytical process, which means that a heuristic must be used. Because of its nature, a heuristic process may not guarantee the absolutely best possible solution, but it will scrutinize the search space to find the most satisfactory solution possible, with the quality of the solutions obtained being dependent on the quality of the heuristic. In this model, after various alternatives were studied and specialists in this area of study were consulted, the authors decided to use a local search heuristic that, because it is relatively straightforward, delivers good results for this type of problem.

As shown in Figure 3, the optimization process starts with a construction algorithm that creates a reasonable initial solution, which is done by use of a greedy algorithm. On the basis of that initial solution, the model runs two cycles: an inner local search process and an outer shaking process. The local search process tries to optimize the solution by searching for better solutions on the search space vicinity of the initial solution, and the shaking process is used to make the solution jump to a different point in the search space to avoid being stuck in a local optimum.

Construction Algorithm

The construction process is based on a greedy algorithm that iteratively improves the links with the highest perceived improvement benefit to their maximum possible level until no more budget for improvements is available. The formula that is used to measure

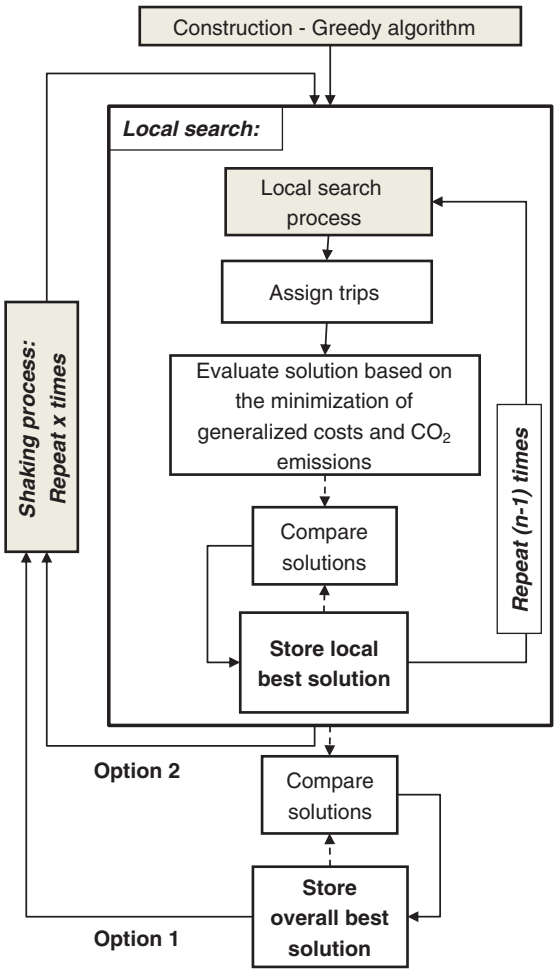


FIGURE 3 Optimization process.

the perceived improvement benefit of each link, which was freely defined by the authors, is the following:

$$IB_i = \left(1 + \frac{h_i}{cap_i}\right)^2 * \sum_{k \in J} f_i^k \quad \forall i \in I, \forall k \in J \tag{18}$$

where

IB_i = improvement benefit of link i,
h_i = maximum one-way traffic flow of all cargo on link i, and
cap_i = capacity (each way) of link i.

As can be seen in Equation 18, the improvement benefit for each link is proportional to the volume of traffic that uses the link and to the relative level of use of the link. The authors considered that links with higher volumes of traffic should be improved first, because their improvement would benefit a larger share of the total traffic. The relative level of use of each link is also important, as links that are over their capacity will benefit the most with a capacity increase, which will allow them to be used by more traffic. If an improvable link has no traffic passing through it, which is the case for links that may be built, the model attributes to it an improvement benefit marginally greater than zero, as an improvement to such a link may be beneficial. The algorithm iteratively improves the links with the

higher value of improvement benefit to their maximum level until no more budget with which to make new improvements is available. The result is an initial network optimization solution.

Local Search Algorithm

The algorithm that is used for the local search process, which is the core of the whole optimization process, is shown schematically in Figure 4.

The local process takes an initial network improvement solution and makes a small change to it by proposing a new solution in the search space vicinity of the initial solution. The process does this by improving at random an improvable link to its optimum level and then iteratively reversing a link at random by one level until the investment is within budget. Each link's optimum level corresponds to the best possible level that a link can reach, without unnecessary improvement to its capacity. This condition means that the optimum level for a link that does not need more capacity is the best possible link level within its capacity bracket. The optimum level for a link that needs more capacity is the best possible link level in the next capacity bracket. The new solution is then tested to see if it is better than the current solution, in which case it becomes the new current solution, and this cycle is repeated as many times as necessary, as defined by the user, as can be seen in Figure 3.

Shaking Algorithm

To avoid the possibility that the model will be stuck in a local optimum, the model includes a shaking algorithm that makes solutions jump to a different place in the search space from which a new local search process can be performed. This process consists of the reversal of a part of the improvement operations that were originally done, followed by the iterative improvement of random links to their optimum level, until the budget is reached or exceeded. An improvement operation corresponds to the improvement of one link by one level, with the relation between the number of reversal operations and the total number of original improvement operations being something that may be calibrated according to the type and the size of the network under analysis. This process creates a new random solution that is significantly different from the original solution and that will likely have left its original search space vicinity to avoid being stuck in local optima. Because the shaking algorithm stops only when the budget is reached or exceeded, most of the improvement solutions from this algorithm will exceed the available budget, at least slightly. This is not a problem, however, because the ensuing local search process will correct this by reversing as many links as needed until the total network investment is within budget.

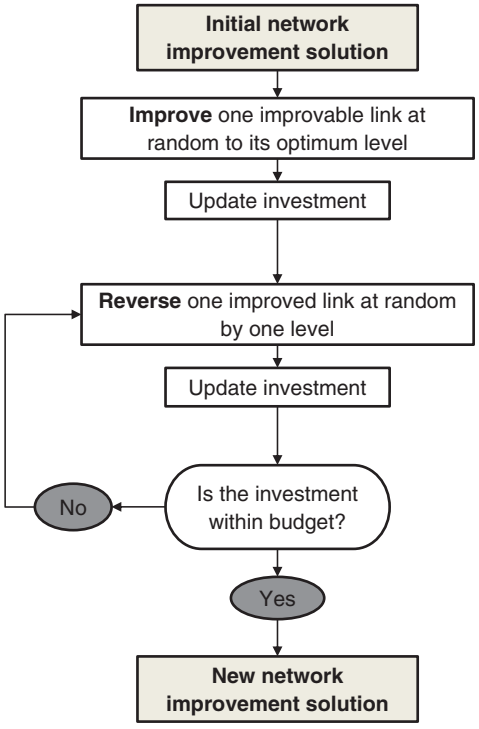


FIGURE 4 Local search process.

As can be seen in Figure 3, two possible options were considered for the initial solution of each shaking cycle: either use of the overall best solution obtained so far (Option 1) or use of the solution obtained in the previous local search process (Option 2). Although the first option has the advantage that it works with the best possible initial solutions, the second allows a greater diversity in those solutions.

Another important aspect of the shaking process that deserves careful consideration is the number of reversal operations performed in each process, which defines the magnitude of change caused by the process. In light of the possible alternatives, the authors tested the model using the two options considered for the initial solution of the shaking process, as well as using different numbers of reversal operations that varied from one-third to two-thirds of the total original improvement operations of each solution. After performing multiple tests focusing on these subjects, the authors concluded that use of the overall best solution at the beginning of each shaking cycle (Option 1) and reversing one-third of the improved links was the technique that presented the best results for the shaking algorithm, and this was therefore the solution adopted.

The superiority of the chosen solution can be perceived in Table 1, which compares the results obtained by the optimization program through the use of different solutions for the shaking algorithm. These values are the result of the application of the model to a

TABLE 1 Comparison of Solutions for Shaking Process

Percentage of Weighted Change in GC and EI	Number of Occurrences by Solution		Percentage of Change
	Alternative[a]	Adopted[b]	
0 < change < +2	20	15	−25.0
−2 < change < 0	537	254	−52.7
−4 < change < −2	970	695	−28.4
−6 < change < −4	264	271	2.7
−8 < change < −6	509	819	60.9
−10 < change < −8	178	348	95.5
−12 < change < −10	22	98	345.5
Total	2,500	2,500	0

[a]Option 2 for initial solution of each shaking cycle and reversing ⅔ of the improved links.
[b]Option 1 for initial solution of each shaking cycle and reversing ⅓ of the improved links.

specific network (Network A, used for the application of the model) comprising 50 shaking cycles with 50 local search cycles, for a total of 2,500 iterations. Each option adopted was superior to the alternative in regard to either the initial solution for the shaking cycle or the number of reversed links because the combination of both was clearly superior to the combination of the alternatives. This can be appreciated by the higher number of iterations with good results (a greater reduction in the weighted total generalized cost and environmental impact) and the consequent lower number of iterations with less satisfactory results.

APPLICATION OF MODEL

Network Descriptions and Scenarios Considered

To test and evaluate the performance of the model developed, the model was applied to two networks created by the authors, which are schematized in Figures 5 and 6. These are relatively simple networks with six traffic-generating poles (centroids), represented as large green dots; road and rail links, which are plain blue lines and red lines with crosses–dashes, respectively; and intermodal terminals, represented in orange. Although the two networks feature the same road network, they feature considerable differences in the rail networks, with Network A being more complex because it includes a congested rail node. The authors opted to run the model on two rail networks to test its adaptability to different conditions.

Network A has a concentration of nodes in the convergence of Links 20, 21, 22, 23, and 26, which represents a congested rail node, as exemplified in Figure 2, in which the virtual link is Link 38. Network B features a simpler rail network without congested nodes, but it offers the possibility for all centroids to be connected by rail. Each link type has various possible link levels, as can be seen in Table 2, with only some of them being used in Networks A and B.

Figures 5 and 6 show some rail links as dashed lines, which means that they are Link Level 0, which represents just the possibility that a link will be built. All the other rail links are either Level 4 (double line) or Level 2 (single line). The intermodal terminals as well as the virtual link representing the congested rail node are all Level 1. The origin–destination matrices for both networks with the demand for freight between the six centroids as well as the movement of passenger trains, which are all model inputs, are presented in Table 3.

The values that were considered for the network improvement costs, the attributes of the different links levels, and the origin–destination matrices were chosen by the authors for this specific application to be realistic indicative values. For simplification, the link attributes for both intermodal cargo and generalized cargo were considered equal, with both types of cargo being measured in tons. CO_2 emissions were quantified as the number of grams per kilometer on the road and rail links and the number of grams per ton of cargo moved in intermodal terminals. For the links capacity, in the case of intermodal terminals it was measured as the number of tons of cargo moved, and in rail links and congested rail nodes it was measured as the number of trains.

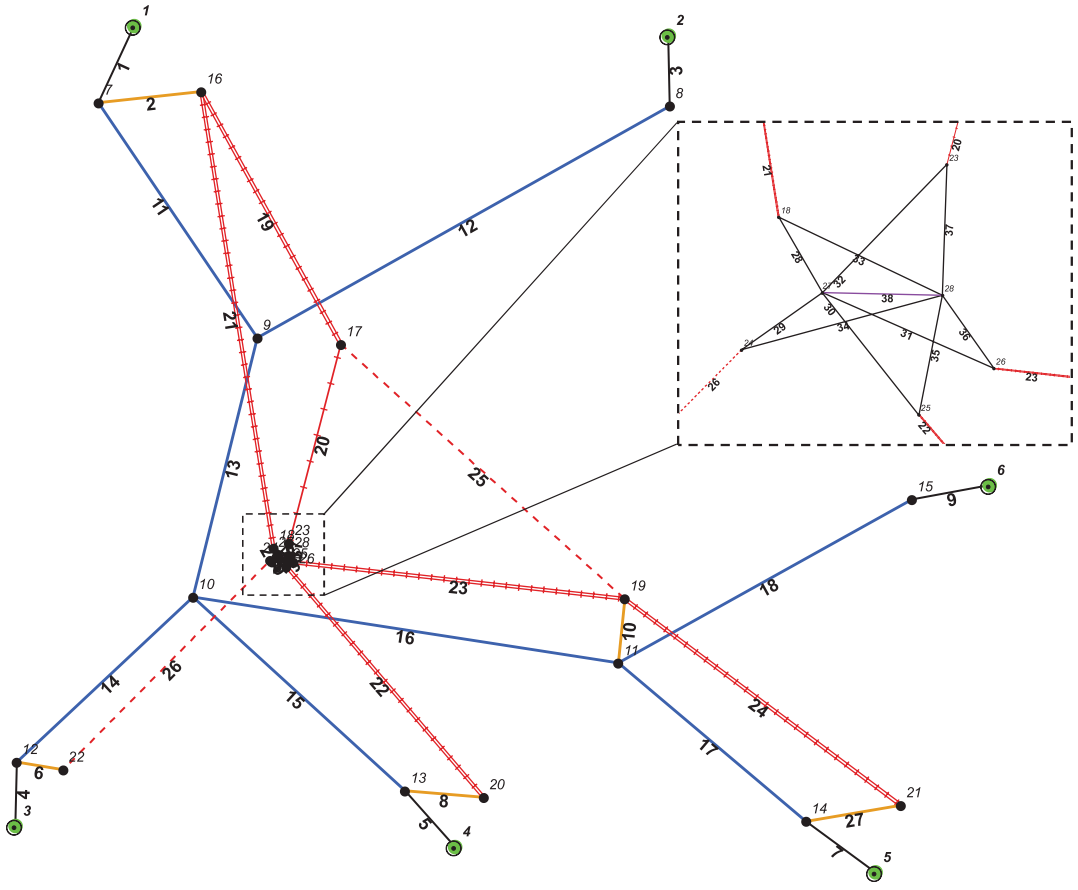


FIGURE 5 Map of Network A.

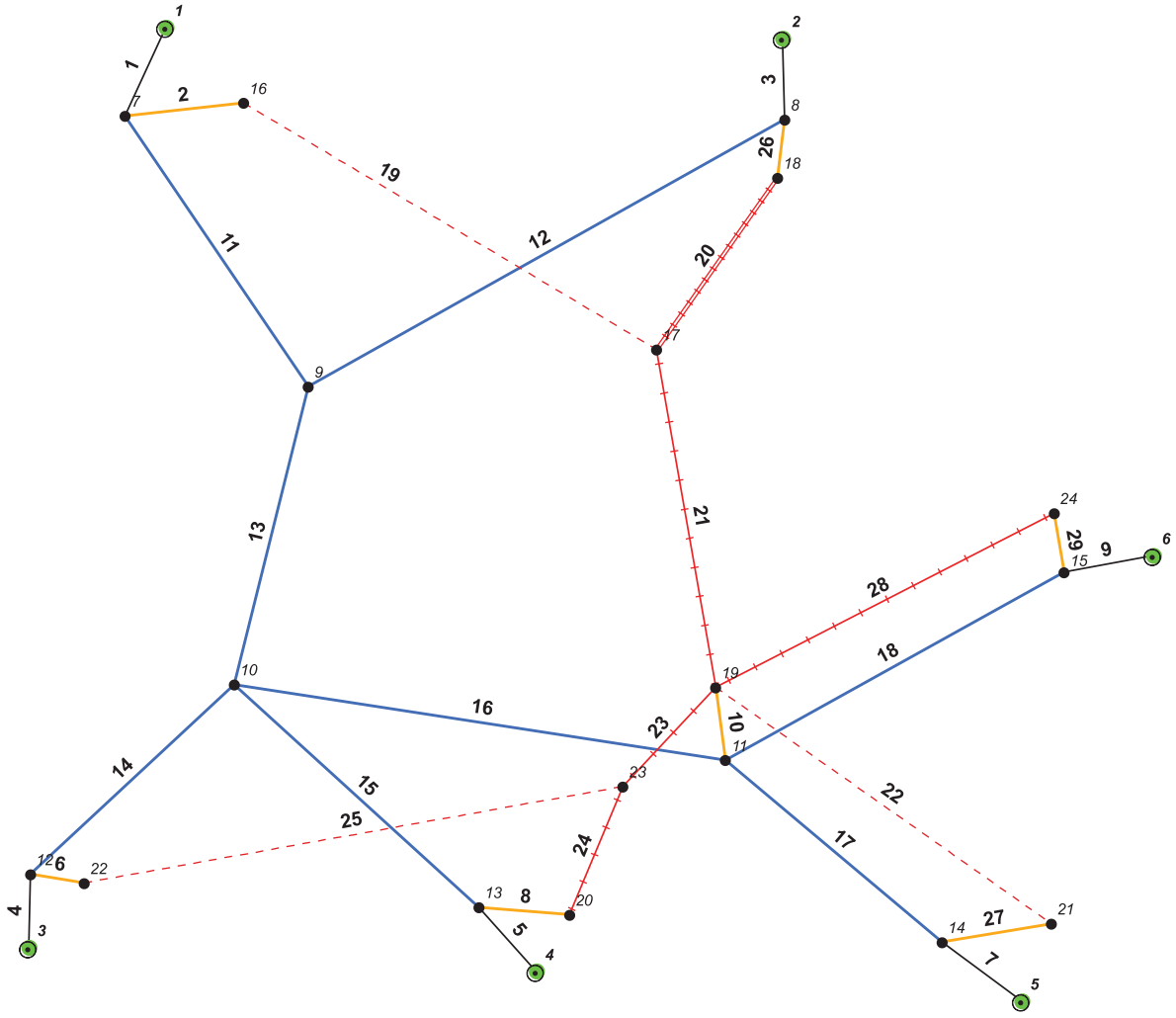


FIGURE 6 Map of Network B.

TABLE 2 Summary of Link Characteristics

Link Level	Type 0: Connector	Type 1: Road Link	Type 2: Rail Link	Type 3: Intermodal Terminal	Type 4: Rail Node Virtual Link
0	—	—	Possible link	Possible link	—
1	Centroid to rail	Road link	Nonelectrified single line; max. train length = 450 m	Intermodal terminal: Capacity Level 1	Rail node: Capacity Level 1
2	Centroid to road	—	Electrified single line; max. train length = 450 m	Intermodal terminal: Capacity Level 2	Rail node: Capacity Level 2
3	Port to rail	—	Electrified single line; max. train length = 750 m	Intermodal terminal: Capacity Level 3	Rail node: Capacity Level 3
4	Port to road	—	Electrified double line; max. train length = 450 m	Intermodal terminal: Capacity Level 4	Rail node: Capacity Level 4
5	Zero-cost connector for rail nodes	—	Electrified double line; max. train length = 750 m	—	Rail node: Capacity Level 5
6	—	—	—	—	Rail node: Capacity Level 6

NOTE: — = no data; max. = maximum.

TABLE 3 Origin–Destination Matrices

Origin–Destination	General Cargo (tons)/Intermodal Cargo (tons)/Passenger Trains (number of trains) by Centroid					
	1	2	3	4	5	6
1	0/0/0	0/0/0	40,000/0/0	25,000/0/7	22,500/55,000/5	35,000/0/0
2	0/0/0	0/0/0	30,000/0/0	20,000/0/0	27,500/0/0	15,000/0/0
3	40,000/0/0	30,000/0/0	0/0/0	0/0/0	0/0/0	0/0/0
4	25,000/0/7	20,000/0/0	0/0/0	0/0/0	0/0/0	0/0/0
5	22,500/55,000/5	27,500/0/0	0/0/0	0/0/0	0/0/0	0/0/0
6	35,000/0/0	15,000/0/0	0/0/0	0/0/0	0/0/0	0/0/0

Application Results and Discussion of Results

The network optimization model was applied to both networks under the following conditions: for the assignment process, the traffic was introduced into the network in 20 interactions; for the optimization process, each local search process consisted of 50 cycles and the shaking process considered 50 shaking cycles. The optimization program took approximately 17 and 10 min to finish with a dual-core 2.5-GHz processor for Networks A and B, respectively. These are reasonable amounts of time for networks of this size and with this rather large number of improvement possibilities. The relative weights that were given to generalized costs and CO_2 emissions minimization were 2 and 1, respectively, and two scenarios were considered: one with a total available budget of 250 million monetary units and another with a total available budget of 500 million monetary units.

The results and improvement solutions obtained for both networks and for each scenario, as well as the type and original level of each link, are presented in Tables 4 and 5. The solutions obtained for each of the two scenarios are considerably different between Networks A and B, a finding that reflects the complexity of the network optimization process, as a bigger budget allows more ambitious network interventions.

For Network A, in the solution obtained for Scenario 1, Virtual Link 38, which represents the congested rail node, is improved to have a higher capacity. This improvement is justified by the congestion of this node in the original network configuration. Rail Links 19 and 20 are also improved to their best possible level to reduce rail transport costs. In contrast, in the solution obtained for Scenario 2, the rail node link is not improved. This is justified because the higher available budget allowed the construction of Link 25, which is a new rail link that diverts rail traffic from the congested node. The diverted traffic means that it no longer needs a capacity improvement. The construction of this new rail link, combined with the improvement of Rail Links 19 and 24, makes the rail mode much more competitive for certain routes. This causes a sharp rise in the amount of cargo that uses Intermodal Terminals 2 and 27, which therefore need to be improved to accommodate this traffic growth.

For the results obtained for Network B, the solution adopted for Scenario 1 includes improvement of Rail Links 20, 21, and 28 and the construction of Link 22. These improvements make the rail mode more competitive because they lower its costs, but because Link 21 did not need a capacity improvement indicates that the amount of traffic using it is still not large. The higher available budget in Scenario 2 allows the implementation of a bolder solution, which includes the construction of Link 19 and the improvement of both capacity and costs for Link 21. With the connection of Centroid 1 with the rail network, this solution generates a significant increase in rail traffic, which justifies the improvement of Link 21, as well as the improvement of Intermodal Terminals 2 and 10.

The results obtained were in line with what was expected, with the model demonstrating the adaptability needed to handle a problem as complex as this optimization problem. This adaptability is patent in the different outcomes that were obtained for the two scenarios in both networks and that occurred because the larger budget of Scenario 2 enabled the adoption of a radically different solution.

CONCLUSIONS

This paper presents an innovative network optimization model that was subsequently applied to two networks under different scenarios. The model is conceived for use at a strategic level of planning for

TABLE 4 Results of Optimization Process

Scenario	Total Investment (million monetary units)	Percentage of Reduction in GC	Percentage of Change in Total CO_2 Emissions	Percentage of Weighted Change in GC and EI
Network A				
1	247	–0.1390	–1.9433	–0.7404
2	484	–3.5632	–23.8656	–10.3306
Network B				
1	249	–0.7633	–5.9569	–2.4945
2	496	–0.3532	–26.5079	–9.0714

TABLE 5 Improvement Solution

ID	Scenario 1				Scenario 2			
	Link Type	Original Link Status	Improved Link Status	Levels of Improvement	Link Type	Original Link Status	Improved Link Status	Levels of Improvement
Network A								
1	0	2	2	0	0	2	2	0
2	3	1	1	0	3	1	3	2
3	0	2	2	0	0	2	2	0
4	0	2	2	0	0	2	2	0
5	0	2	2	0	0	2	2	0
6	3	1	1	0	3	1	1	0
7	0	2	2	0	0	2	2	0
8	3	1	1	0	3	1	1	0
9	0	2	2	0	0	2	2	0
10	3	1	1	0	3	1	1	0
11	1	1	1	0	1	1	1	0
12	1	1	1	0	1	1	1	0
13	1	1	1	0	1	1	1	0
14	1	1	1	0	1	1	1	0
15	1	1	1	0	1	1	1	0
16	1	1	1	0	1	1	1	0
17	1	1	1	0	1	1	1	0
18	1	1	1	0	1	1	1	0
19	2	4	5	1	2	4	5	1
20	2	2	5	3	2	2	2	0
21	2	4	4	0	2	4	4	0
22	2	4	4	0	2	4	4	0
23	2	4	4	0	2	4	4	0
24	2	4	4	0	2	4	5	1
25	2	0	0	0	2	0	5	5
26	2	0	0	0	2	0	0	0
27	3	1	1	0	3	1	3	2
28	0	5	5	0	0	5	5	0
29	0	5	5	0	0	5	5	0
30	0	5	5	0	0	5	5	0
31	0	5	5	0	0	5	5	0
32	0	5	5	0	0	5	5	0
33	0	5	5	0	0	5	5	0
34	0	5	5	0	0	5	5	0
35	0	5	5	0	0	5	5	0
36	0	5	5	0	0	5	5	0
37	0	5	5	0	0	5	5	0
38	4	1	2	1	4	1	1	0

(continued)

TABLE 5 *(continued)* **Improvement Solution**

	Scenario 1				Scenario 2			
ID	Link Type	Original Link Status	Improved Link Status	Levels of Improvement	Link Type	Original Link Status	Improved Link Status	Levels of Improvement
Network B								
1	0	2	2	0	0	2	2	0
2	3	1	1	0	3	1	3	2
3	0	2	2	0	0	2	2	0
4	0	2	2	0	0	2	2	0
5	0	2	2	0	0	2	2	0
6	3	1	1	0	3	1	1	0
7	0	2	2	0	0	2	2	0
8	3	1	1	0	3	1	1	0
9	0	2	2	0	0	2	2	0
10	3	1	1	0	3	1	3	2
11	1	1	1	0	1	1	1	0
12	1	1	1	0	1	1	1	0
13	1	1	1	0	1	1	1	0
14	1	1	1	0	1	1	1	0
15	1	1	1	0	1	1	1	0
16	1	1	1	0	1	1	1	0
17	1	1	1	0	1	1	1	0
18	1	1	1	0	1	1	1	0
19	2	0	0	0	2	0	5	5
20	2	4	5	1	2	4	4	0
21	2	2	3	1	2	2	5	3
22	2	0	3	3	2	0	0	0
23	2	2	2	0	2	2	2	0
24	2	2	2	0	2	2	2	0
25	2	0	0	0	2	0	0	0
26	3	1	1	0	3	1	1	0
27	3	1	1	0	3	1	1	0
28	2	2	3	1	2	2	2	0
29	3	1	1	0	3	1	1	0

NOTE: ID = link number.

modeling of major road and rail links, as well as specific congested rail nodes and intermodal terminals. It is an innovative model because it is not limited, unlike existing freight network optimization models, which are usually limited because they allow only the construction or the improvement of links or because they have a limited search space. This model allows both the improvement of existing links and the construction of new ones from scratch and can be applied to different networks without a limit on the number or variety of network improvement possibilities.

The practical application of the model produced satisfactory results, which highlight its versatility, as it was able to optimize investments in different networks through the consideration of two investment scenarios and to deliver diverse solutions adapted for each situation.

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REFERENCES

1. Maia, L. C., and A. F. D. Couto. Development of a Strategic Freight Network Optimization Model. *Proc., XXV ANPET—Congresso de Pesquisa e Ensino em Transportes,* Belo Horizonte, Brazil, 2011, pp. 443–453.
2. Crainic, T. G., and G. Laporte. Planning Models for Freight Transportation. *European Journal of Operational Research,* Vol. 97, No. 3, 1997, pp. 409–438.
3. Wigan, M. R., and F. Southworth. What's Wrong with Freight Models, and What Should We Do About It? Presented at 85th Annual Meeting of the Transportation Research Board, Washington, D.C., 2006.
4. Jourquin, B. A Multi-Flow Multi-Modal Assignment Procedure Applied to the European Freight Transportation Networks. *Studies in Regional Science,* Vol. 35, No. 4, 2005, pp. 929–945.
5. Crainic, T. G., M. Florian, and J.-E. Léal. A Model for the Strategic Planning of National Freight Transportation by Rail. *Transportation Science,* Vol. 24, No. 1, 1990, pp. 1–24.
6. Guélat, J., M. Florian, and T. G. Crainic. A Multimode Multiproduct Network Assignment Model for Strategic Planning of Freight Flows. *Transportation Science,* Vol. 24, No. 1, 1990, pp. 25–39.
7. Jourquin, B., and M. Beuthe. Transportation Policy Analysis with a Geographic Information System: The Virtual Network of Freight Transportation in Europe. *Transportation Research Part C,* Vol. 4, No. 6, 1996, pp. 359–371.
8. Beuthe, M., B. Jourquin, J.-F. Geerts, C. Koul, and N. Ha. Freight Transportation Demand Elasticities: A Geographic Multimodal Transportation Network Analysis. *Transportation Research Part E,* Vol. 37, No. 4, 2001, pp. 253–266.
9. Abdelwahab, W. M. Elasticities of Mode Choice Probabilities and Market Elasticities of Demand: Evidence from a Simultaneous Mode Choice/Shipment-Size Freight Transport Model. *Transportation Research Part E,* Vol. 34, No. 4, 1998, pp. 257–266.
10. Shinghal, N., and T. Fowkes. Freight Mode Choice and Adaptive Stated Preferences. *Transportation Research Part E,* Vol. 38, No. 5, 2002, pp. 367–378.
11. Andersen, J., and M. Christiansen. Designing New European Rail Freight Services. *Journal of the Operational Research Society,* Vol. 60, No. 3, 2009, pp. 348–360.
12. Vanek, F. M., and R. I. Smith. Prospects for Rail Freight from Peripheral Regions: The Case of North-East Scotland and Grampian Country Foods. *International Journal of Logistics,* Vol. 7, No. 1, 2004, pp. 59–70.
13. Oum, T. H. Derived Demand for Freight Transport and Inter-Modal Competition in Canada. *Journal of Transport Economics and Policy,* Vol. 13, No. 2, pp. 149–168.
14. Tsamboulas, D. A., and P. Moraitis. Methodology for Estimating Freight Volume Shift in an International Intermodal Corridor. In *Transportation Research Record: Journal of the Transportation Research Board, No. 2008,* Transportation Research Board of the National Academies, Washington, D.C., 2007, pp. 10–18.
15. Yang, L., Z. Gao, and K. Li. Railway Freight Transportation Planning with Mixed Uncertainty of Randomness and Fuzziness. *Applied Soft Computing,* Vol. 11, No. 1, 2011, pp. 778–792.
16. Santos, B. F., A. P. Antunes, and E. J. Miller. Interurban Road Network Planning Model with Accessibility and Robustness Objectives. *Transportation Planning and Technology,* Vol. 33, No. 3, 2010, pp. 297–313.
17. Yamada, T., B. F. Russ, J. Castro, and E. Taniguchi. Designing Multimodal Freight Transport Networks: A Heuristic Approach and Applications. *Transportation Science,* Vol. 43, No. 2, 2009, pp. 129–143.
18. Arnold, P., D. Peeters, and I. Thomas. Modelling a Rail/Road Intermodal Transportation System. *Transportation Research Part E,* Vol. 40, No. 3, 2004, pp. 255–270.
19. Zhang, G., and J. Lu. Genetic Algorithm for Continuous Network Design Problem. *Journal of Transportation Systems Engineering and Information Technology,* Vol. 7, No. 1, 2008, pp. 101–105.
20. Crainic, T. G. Service Network Design in Freight Transportation. *European Journal of Operational Research,* Vol. 122, No. 2, 2000, pp. 272–288.
21. Yaghini, M., A. Foroughi, and B. Nadjari. Solving Railroad Blocking Problem Using Ant Colony Optimization Algorithm. *Applied Mathematical Modelling,* Vol. 35, No. 12, 2011, pp. 5579–5591.
22. Maia, L. C., and A. F. D. Couto. An Innovative Freight Traffic Assignment Model for Multimodal Networks. *Computers in Industry,* Vol. 64, No. 2, 2013, pp. 121–127.
23. Floyd, R. W. Algorithm 97: Shortest Path. *Communications of ACM,* Vol. 5, 1962, p. 345.

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